Metagenomic preparation results

Complete statistics and explanations | Preparation version 2.0.0 | 17 October 2026

Preparation completed successfully. The main dataset contains **36,590 QC-passed runs and 3,425 GMrepo-reported abundance features**.

The uploaded summary reports status **complete**. Its counts reconcile, its source hashes match the earlier extraction audit, and its script hash matches the delivered script.

Run and QC counts

QC category	With usable profile	Without profile	Total runs
QC passed	**36,590**	**7,771**	**44,361**
QC failed	**0**	**6,600**	**6,600**
QC missing	**0**	**5,308**	**5,308**
Total	**36,590**	**19,679**	**56,269**

56,269 metagenomic runs are retained in the master metadata. **44,361** passed QC, but **7,771** of these lacked species profiles.

36,590 runs (65.03% of the extracted inventory) entered the abundance matrix. No QC-conflict or unrecognized-QC category was recorded.

The exclusion reasons overlap. Do not add the three terminal counts: the 6,600 QC-failed and 5,308 QC-missing runs are already among the 19,679 runs without species profiles.

ASD annotation counts

Annotation group	Original inventory	Included in matrix	Outside matrix
ASD annotation observed	**252**	**208**	**44**
No ASD annotation observed	**56,017**	**36,382**	**19,635**
Ambiguous ASD annotation	**0**	**0**	**0**
Total	**56,269**	**36,590**	**19,679**

The proposed non-ASD pretraining subset has **36,382 runs**; the ASD subset has **208 runs**. These groups describe annotations. **The non-ASD subset is not necessarily a healthy-control dataset.**

These are sequencing-run counts, not confirmed counts of distinct participants. Their biological grouping can be traced through the preserved metadata.

Feature and output-file statistics

Observed preparation results | The following pages explain how to interpret these numbers.

Statistic	Count
Original abundance records examined	**3,198,365**
Original catalogue entries, including name variants and placeholder	**3,436**
Placeholder/invalid-ID catalogue entry excluded	**1**
Final distinct feature columns	**3,425**
Features with multiple supplied name variants	**10**
Proven repeated measurements collapsed	**3,710**
Named features omitted for having no positive values in the main matrix	**0**

Two supplied names occurred under different TaxIDs, producing **four ID-suffixed headers** to preserve separate features: Lacrimispora amygdalina and Microbacterium maritypicum. **No NCBI mapping was performed.**

What the output files contain

File	Data rows	Abundance features	Compressed size
all_metadata.tsv.gz	**56,269**	Metadata and preparation decisions	9.30 MB
qc_pass_abundance.tsv.gz	**36,590**	3,425	20.33 MB
asd_qc_pass_abundance.tsv.gz	**208**	3,425	0.155 MB
non_asd_qc_pass_abundance.tsv.gz	**36,382**	3,425	20.20 MB
qc_fail_abundance_long.tsv.gz	**0 abundance records**	Original long-table header	157 bytes

Each abundance matrix has **3,426 columns altogether**: run_id plus 3,425 abundance columns. Sizes above use decimal MB and are rounded.

The tiny QC-failed file is expected. There are 6,600 QC-failed runs, but GMrepo returned no species-abundance records for them. Their metadata remains in all_metadata.tsv.gz; those runs were not lost.

Abundances remain as reported. **No transformations, dataset splitting, masking or model training has occurred.** These features follow GMrepo species-profile reporting; their current taxonomic ranks were not externally validated.

Review scope: the uploaded filtering_summary.json and its source/script references were checked. The five generated data tables themselves were not uploaded for independent inspection.

Understanding the preparation statistics

Explanation 1 of 2 | Actual counts from your completed preparation, version 2.0.0.

The numbers count different things. A run is a sequencing run. An abundance record is one run-organism measurement. A feature is one organism column in the matrix. Runs are not necessarily distinct people.

A small example

Run	Organism	Abundance
Run 1	Organism A	**10**
Run 1	Organism B	**5**
Run 2	Organism A	**7**

Illustrative values only. This example has **3 records, 2 runs and 2 features**. The same organism can appear in many runs.

Your actual statistics

Statistic	Count	Meaning
Original abundance records	**3,198,365**	Individual run-organism measurements examined before local cleanup. This is not the number of species or participants.
Original catalogue entries	**3,436**	Distinct original ID-and-name combinations. Two supplied spellings for one ID can create two catalogue entries.
Placeholder entry excluded	**1**	The Unknown/invalid-ID catalogue entry was omitted as a feature. This does not mean one whole run was removed.
Features with name variants	**10**	Ten original IDs each had two supplied spellings. One matrix column was used per ID; the spellings remain in the summary.
Repeated records collapsed	**3,710**	Extra, proven repeated measurements within runs were kept once. This is not a count of removed runs or species.
Final feature columns	**3,425**	Distinct retained GMrepo organism IDs, displayed with supplied species names. Their current taxonomic ranks were not externally validated.
Named features omitted for all-zero abundance	**0**	Every retained feature had a positive value in at least one included run. It need not be positive in every run.

Catalogue arithmetic: 3,436 entries - 1 Unknown entry - 10 extra spelling entries = **3,425 feature columns**.

Duplicate example: two proven repeated entries of 50 for the same organism in the same run become **one value of 50**, not 100. Raw-response checks support the collapse.

Sources for both pages: your uploaded filtering_summary.json and prepare_metagenomic_X.py, version 2.0.0. Examples are illustrative. The five generated tables were not independently reread here.

How the shared abundance matrix was built

Explanation 2 of 2 | Union of features and zero filling.

Different runs report different numbers of organisms. Shared columns give every included run an abundance vector of the same length.

The four steps

1. Select eligible QC-passed runs with usable profiles from all phenotype groups.

2. Collect the union of retained named features with positive abundance across these runs. In your data, this gives **3,425 features across 36,590 runs**.

3. Preserve reported abundances. Keep different original TaxIDs as separate features.

4. Write **0** where a retained feature was not reported for that run. Existing reported zero values also remain zero.

An illustrative union example

Run 1 reports A = 10 and C = 5. Run 2 reports B = 8 and C = 3. The union of their features is **A, B and C**.

run_id	A	B	C
Run 1	**10**	**0**	**5**
Run 2	**0**	**8**	**3**

Illustrative values only. The zeros fill entries that were not reported in these example profiles.

Two cases that must be kept separate

An organism is unreported within an available, usable profile: its matrix cell is filled with 0. This means unreported/not detected, rather than proven biological absence.

The entire profile is unavailable: the run is excluded from the abundance matrix and remains in the master metadata. The script does not invent an all-zero row for that run.

Your three matrices

Dataset	Run rows	Abundance columns
All eligible QC-passed groups	**36,590**	**3,425**
ASD annotation observed	**208**	**3,425**
No ASD annotation observed	**36,382**	**3,425**

The ASD and non-ASD subsets select rows from the main matrix. They inherit **the same columns, in the same order, with the same row values**. A feature can be zero throughout one subset while being positive in the other.

Each file has **3,426 columns in total**: run_id plus 3,425 abundance columns. The run_id column is an identifier and is not a numeric model feature.

No rescaling or renormalization was applied. No model has been trained. No ASD annotation does not automatically mean a healthy control.